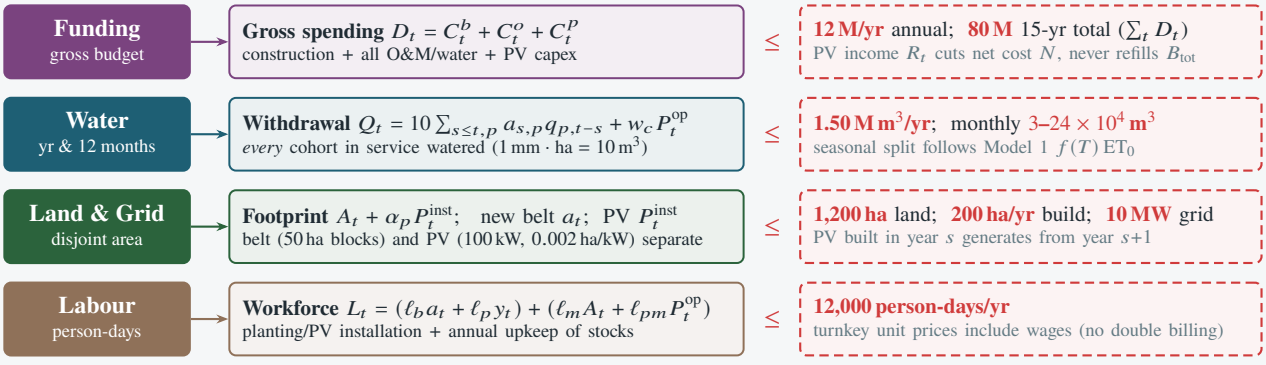

DIMENSION
DEMAND LEDGER (NEW FLOW + ACTIVE STOCK)
SCENARIO CAP & ACCOUNTING RULE
Funding
gross budget
Gross spending $D_t = C_t^b + C_t^o + C_t^p$
construction + all O&M/water + PV capex
≤
12 M/yr annual; 80 M 15-yr total ($\sum_t D_t$)
PV income R_t cuts net cost N, never refills B_{tot}
Water
yr & 12 months
Withdrawal $Q_t = 10 \sum_{s \le t,p} a_{s,p} q_{p,t-s} + w_c P_t^{\text{op}}$
every cohort in service watered (1 mm · ha = 10 m³)
≤
1.50 M m³/yr; monthly 3–24 × 10⁴ m³
seasonal split follows Model 1 $f(T)\,\text{ET}_0$
Land & Grid
disjoint area
Footprint $A_t + \alpha_p P_t^{\text{inst}}$; new belt a_t; PV P_t^{inst}
belt (50 ha blocks) and PV (100 kW, 0.002 ha/kW) separate
≤
1,200 ha land; 200 ha/yr build; 10 MW grid
PV built in year s generates from year s+1
Labour
person-days
Workforce $L_t = (\ell_b a_t + \ell_p y_t) + (\ell_m A_t + \ell_{pm} P_t^{\text{op}})$
planting/PV installation + annual upkeep of stocks
≤
12,000 person-days/yr
turnkey unit prices include wages (no double billing)